

LEARNING MANAGEMENT SYSTEM

Master Technical Architecture, System Engineering & Complete Visual Specification

System Name:	Enterprise Academic Learning Management System (LMS)
Author / Lead:	Abinash Kar
Core Framework:	Spring Boot 4.x / 3.x (Java 17/24 Runtime)
Persistence Layer:	Spring Data JPA & Hibernate 7 ORM
Database:	MySQL 8.0 (Relational InnoDB Schema)
Security Engine:	Spring Security (Role-Based Access Control)
Presentation:	Server-Side JSP, JSTL, Bootstrap 5, Chart.js
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Executive Overview

The **Learning Management System (LMS)** is an enterprise-grade academic platform designed to orchestrate the entire educational lifecycle across three principal user archetypes: **Students**, **Faculty**, and **Administrators**. Built on modern Java and Spring Boot architecture, the system provides high-performance video course delivery, real-time assignment submission and grading, asynchronous email broadcasting, role-based authorization, and interactive timed online examinations.

Core System Capabilities at a Glance:

- **Administrator Control Center:** Real-time KPI analytics, user account approval gating, multi-channel notice broadcasting, department management, and mass email broadcast dispatch.
- **Faculty Course & Exam Hub:** Multimedia video lecture streaming, structured assignment authoring with student grading, real-time doubt resolution forum, and timed MCQ exam creation.
- **Student Learning Experience:** Interactive course browsing and enrollment, chapter-based video streaming with automatic progress persistence, assignment uploads, doubt tickets, and automated timed exams.
- **Enterprise Security:** Role-based authorization filters, session status gating, active user presence tracking, and secure credential handling.

1. System Architecture & Multi-Tier Design

The LMS application is engineered following an enterprise multi-tier Model-View-Controller (MVC) and layered service-oriented architecture. This decoupling guarantees strict separation of concerns, high maintainability, database independence via ORM abstractions, and seamless scalability.

Layer	Components & Technologies	Architectural Responsibility
Presentation Tier	JSP, JSTL, Bootstrap 5, Bootstrap Icons, Chart.js, Vanilla CSS	Renders responsive, glassmorphism-styled UI components, handles client-side form validation, chart visualizations, and video playback.
Controller / Routing Tier	Spring MVC Controllers (@Controller, @RestController)	Intercepts incoming HTTP GET/POST requests, extracts parameters, validates session tokens, enforces role authorization, and routes to views or REST payloads.
Business Service Tier	Spring Services (@Service), EmailService, VideoService	Encapsulates business rules, orchestrates transactions, handles multi-part file uploads, streams video binary streams, and manages asynchronous email queues (@Async).
Persistence & ORM Tier	Spring Data JPA, Hibernate 7, HikariCP Connection Pool	Abstracts relational data access via repository interfaces extending JpaRepository. Executes optimized SQL, manages schema updates (ddl-auto=update), and entity lifecycles.
Database Tier	MySQL 8.0 Engine (InnoDB, UTF-8 Encoding)	Persistent relational database storing users, courses, videos, enrollments, notices, doubts, assignments, exams, submissions, and audit logs.
Security Tier	Spring Security 6.x / Custom Filter Chains	Guarantees Role-Based Access Control (RBAC), intercepting unauthorized route traversal, managing authentication state, and session invalidation on logout.

2. Complete Technology Stack Matrix

Category	Technology / Library	Version	Purpose in LMS Platform
Runtime / Core	Java SE	17 / 24	Primary object-oriented language for business logic and services.
Framework	Spring Boot	4.0.1 / 3.x	Core backend framework offering IoC, DI, autoconfiguration, and embedded Tomcat.
Web Engine	Spring MVC & Embedded Tomcat	11.0.15	HTTP request routing, servlet container, session tracking.
View Engine	JSP & JSTL (Jakarta)	3.0.0	Dynamic server-rendered HTML pages with tags and expression language.
ORM / Data	Spring Data JPA & Hibernate	7.2.0	Object-relational mapping, derived query methods, declarative CRUD.
Database	MySQL Connector/J & MySQL Server	8.0+	Relational database storage with InnoDB ACID transactions.
Security	Spring Security	6.x	RBAC authorization, role-based routing, authentication verification.
Mailing	Spring Boot Starter Mail (JavaMail)	4.0.1	Asynchronous Rich-HTML email delivery via SMTP (Gmail/Custom).
Styling / UI	Bootstrap 5 & Bootstrap Icons	5.3.2	Modern responsive grids, glassmorphism dashboards, modal dialogs.

Charts	Chart.js	4.4.0	Interactive live charts for admin and faculty metric visualizers.
Build Tool	Apache Maven & Wrapper (mvnw)	3.9.x	Dependency resolution, compilation, packaging, and lifecycle management.

3. Role-Based Module & Functional Specification

3.1. Authentication & Security Engine

The security model enforces strict Role-Based Access Control (RBAC). When users register, they are assigned a role (Student, Faculty, or Admin) and placed in a **Pending** state (status = 0). Admins must explicitly approve accounts before login is permitted. Upon successful authentication, the user's session is populated, their online flag (isOnline = 1) is set in real-time, and they are routed to their designated dashboard.

- **Session Governance:** Real-time session attributes (name, email, role) protect routes against unauthorized URL tampering.
- **Live Presence Tracking:** Login sets isOnline = 1 and Logout or Session Invalidation gracefully reverts isOnline = 0.

3.2. Administrator Management Hub (/adashboard)

The Administrator portal is the operational command center of the LMS ecosystem, equipped with real-time statistics, user lifecycle tools, and mass broadcast utilities.

- **KPI Metrics Dashboard:** Visualizes total users, active faculty, enrolled students, active courses, pending doubts, and server health in glassmorphic cards.
- **User Lifecycle Management (/users):** Complete CRUD capability to search, inspect, activate (toggle status 0 ↔ 1), edit roles, or delete users.
- **Multi-Channel Notice Board (/addnotice):** Publish institutional announcements targeted to 'All', 'Student', or 'Faculty' with optional file attachments stored securely in the uploads directory.
- **Asynchronous Broadcast Email Engine (/admin/broadcast):** Dispatch Rich-HTML broadcast emails to segmented cohorts. Offloaded to background worker threads using Spring's @Async JavaMailSender and tracked in a permanent broadcast audit log.
- **Academic Oversight:** Full administrative control over departments, courses, faculty lecture uploads, student doubts, and live exam monitoring.

3.3. Faculty Educational Hub (/fdashboard)

Designed specifically for instructors and professors to author curriculums, deliver multimedia lectures, interact with learners, and assess student performance.

- **Course & Curriculum Management:** Organize syllabi into structured modules and courses mapped to academic departments.
- **Video Lecture Streaming System (/faculty/videos):** Upload full HD video lectures (MP4, MKV, WebM up to 500MB) with auto-generated descriptions, duration metadata, and chapter tags.
- **Assignment Authoring & Grading Portal (/f-assignments):** Post assignments with deadlines, attachments, and max points. Review student submission files, assign grades, and provide feedback.
- **Real-Time Doubt Forum (/faculty/doubts):** Direct Q&A; portal allowing faculty to resolve questions submitted by enrolled students with threaded replies.
- **Online Exam Creator (/faculty/exams):** Build timed MCQ examinations, configure duration in minutes, pass percentage, and author multi-choice questions with answer keys.

3.4. Student Learning Portal (/sdashboard)

A learner-centric single page application (SPA)-like experience designed to accelerate knowledge acquisition and academic tracking.

- **Personalized Learning Dashboard:** Instant summary of enrolled courses, upcoming deadlines, exam alerts, recent notices, and course completion progress bars.
- **Course Catalog & Enrollment (/s-courses, /s-browse-courses):** Explore courses by department, view syllabus, and enroll with one click.
- **Interactive Video Learning Suite (/s-watch):** HTML5 video player with chapter navigation, playback controls, and automated progress calculation.
- **Assignment Submission Hub (/s-assignments):** Download assignment briefs, upload completed solution documents (PDF, DOCX, ZIP), and view instructor grades and remarks.
- **Doubt Ticket Submission:** Submit technical or conceptual queries directly linked to specific courses and receive faculty responses.
- **Online Examination & Assessment Engine (/student/exams):** Take timed exams in a focused portal with countdown timer, dynamic question navigation, instant score calculation, and detailed answer review.
- **Profile & Credential Management (/s-profile):** Update personal contact info, change passwords, and view academic activity logs.

4. Database Architecture & JPA Entities

The relational database schema is modeled with 17 Spring Data JPA entities, utilizing foreign key constraints, indexing on lookup fields, and cascaded operations where appropriate.

Entity	Primary Key	Key Fields & Attributes	Relationships & Foreign Keys
User	id (INT, PK)	name, email (Unique), mobile, password, role, status (0/1), isOnline (0/1)	1:M with Course (Faculty), 1:M with Enrollment, 1:M with Doubt
Course	id (INT, PK)	title, code, description, department, duration, facultyName, createdAt	M:1 with Department, 1:M with VideoLecture, 1:M with Assignment, 1:M with Enrollment
Department	id (INT, PK)	name, code, description, headOfDepartment	1:M with Course
VideoLecture	id (INT, PK)	title, description, videoUrl, videoFileName, durationMinutes, courseId, uploadedBy, createdAt	M:1 with Course
Enrollment	id (INT, PK)	studentEmail, studentName, courseId, courseName, progressPercent, enrolledAt	M:1 with User (student), M:1 with Course
Assignment	id (INT, PK)	title, description, courseId, courseName, dueDate, maxMarks, attachmentPath, createdBy	M:1 with Course, 1:M with AssignmentSubmission
AssignmentSubmission	id (INT, PK)	assignmentId, studentEmail, studentName, submissionText, filePath, marksAwarded, feedback, status, submittedAt	M:1 with Assignment, M:1 with User
Notice	id (INT, PK)	title, content, targetRole (All/Student/Faculty), priority, attachmentPath, postedBy, createdAt	Independent announcement entity
Doubt	id (INT, PK)	studentEmail, studentName, courseId, courseName, question, answer, status (OPEN/RESOLVED), answeredBy, createdAt	M:1 with Course, M:1 with User
Exam	id (INT, PK)	title, courseId, courseName, durationMinutes, totalMarks, passMarks, status (ACTIVE/INACTIVE), createdBy	M:1 with Course, 1:M with Question, 1:M with ExamResult
Question	id (INT, PK)	examId, questionText, marks, questionType	M:1 with Exam, 1:M with Option
Option	id (INT, PK)	questionId, optionText, isCorrect (BOOLEAN)	M:1 with Question
ExamResult	id (INT, PK)	examId, studentEmail, studentName, score, totalMarks, percentage, status (PASSED/FAILED), completedAt	M:1 with Exam, M:1 with User
StudentAnswer	id (INT, PK)	examResultId, questionId, selectedOptionId, isCorrect	M:1 with ExamResult, M:1 with Question
BroadcastLog	id (INT, PK)	subject, recipientCount, targetRole, senderEmail, sentAt, status	Audit entity for mass email tracking
FAQ	id (INT, PK)	question, answer, category, role (All/Student/Faculty)	Dynamic institutional knowledge base
Contact	id (INT, PK)	name, email, mobile, subject, message, submittedAt	Public inquiries & contact messages

5. Visual Showcase & User Interface Architecture

Below is a comprehensive visual walkthrough of every major user interface, detailing screen purpose, responsive design paradigms, and user interactions.

Exhibit 1: LMS Landing Page (/ & /index)

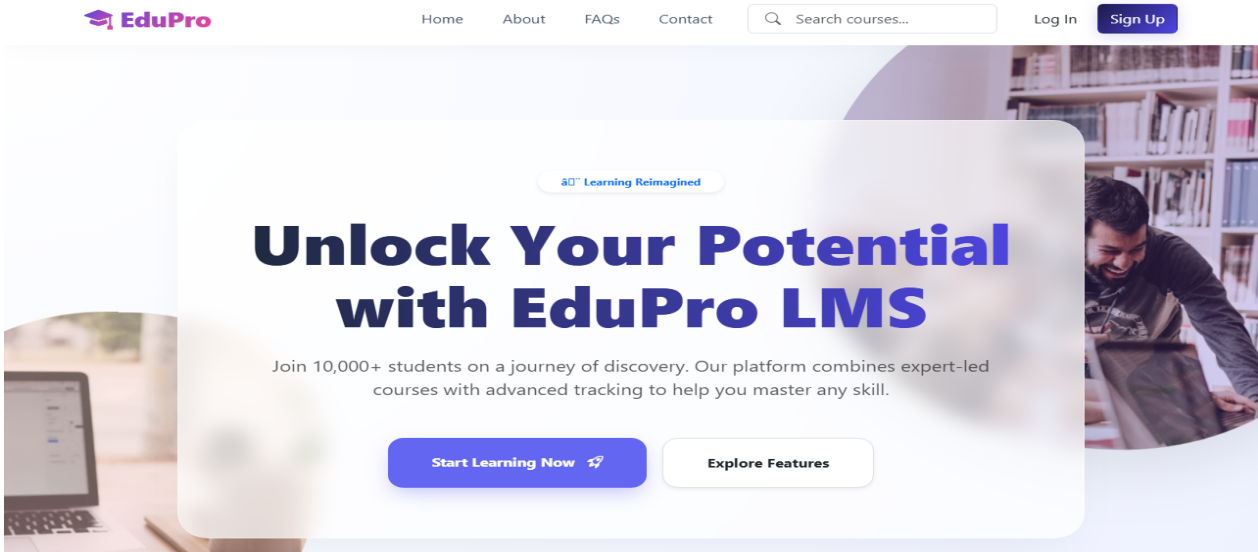


Figure 1: Public portal featuring hero banners, quick navigation links, platform statistics, course teasers, and authentication entry points.

Exhibit 2: Unified Authentication Portal (/login)

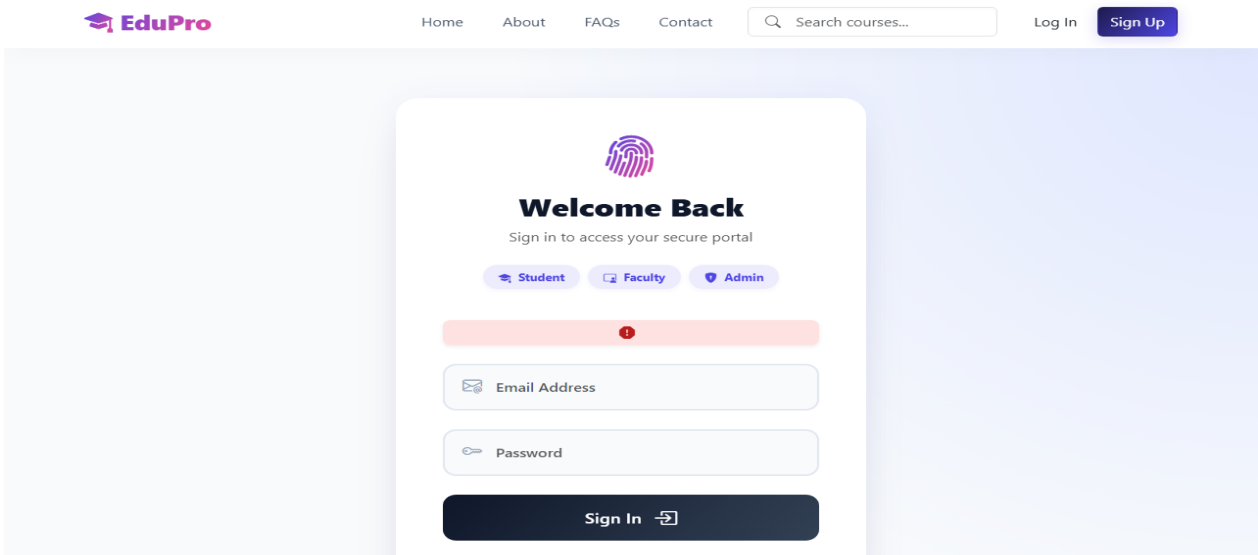


Figure 2: Secure login screen with role auto-detection, password verification, active account status checks, and self-registration redirect.

Exhibit 3: Administrator Control Center & Analytics Dashboard (/adashboard)

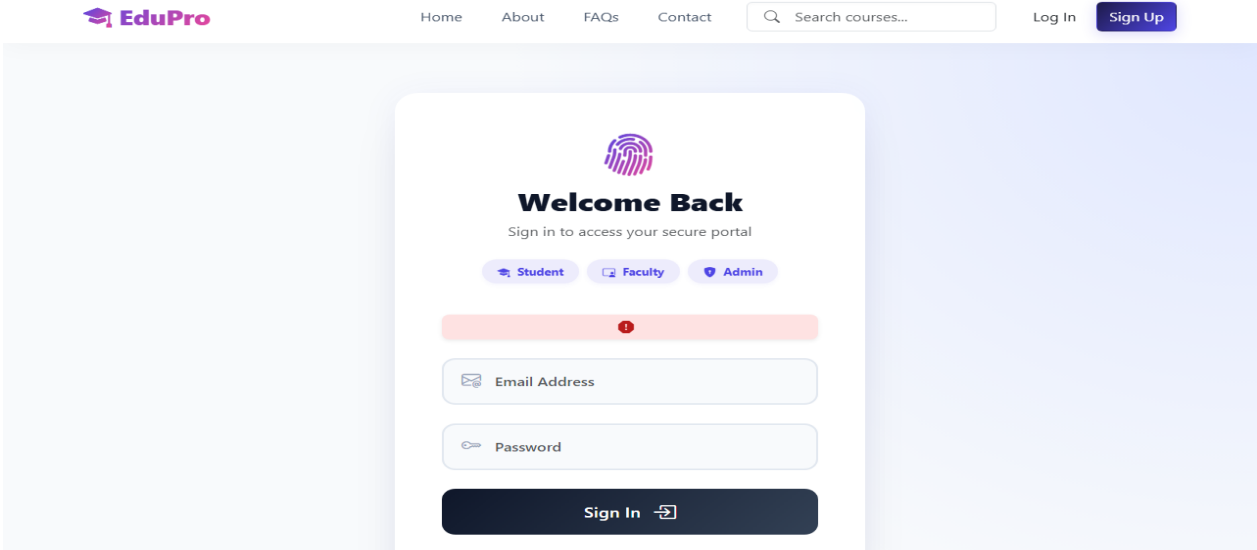


Figure 3: Glassmorphism executive dashboard with dynamic KPI cards, student vs faculty metrics, active sessions, and system navigation hub.

Exhibit 4: Enterprise User Lifecycle Management (/users)

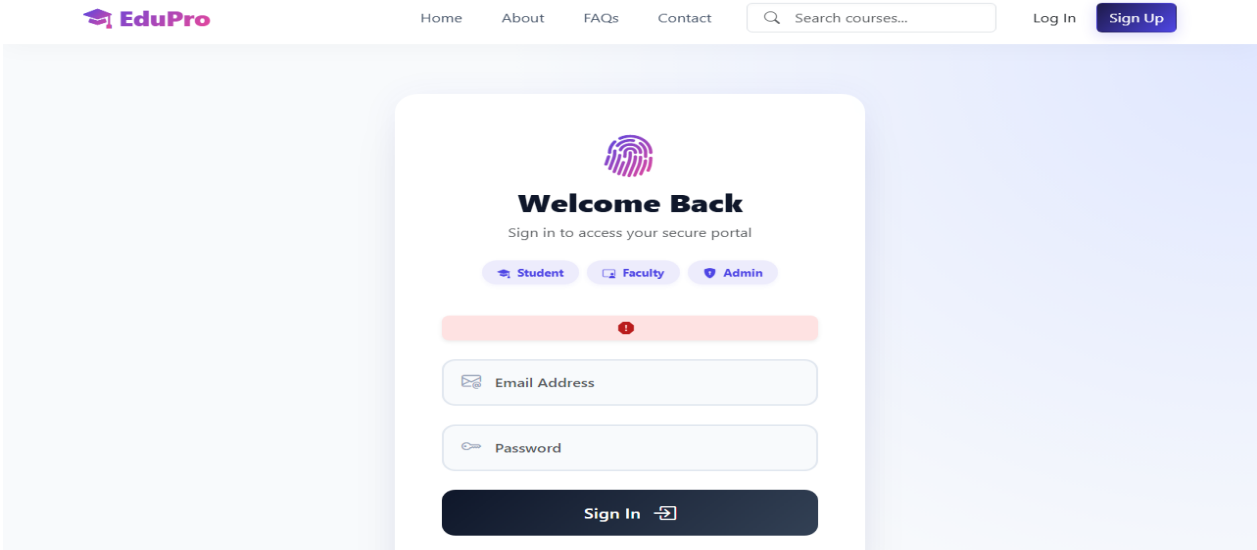


Figure 4: Tabular user directory supporting instant account activation (status 0 to 1 toggle), role edits, search filtering, and deletion.

Exhibit 5: Institutional Announcement & Notice Dispatcher (/addnotice)

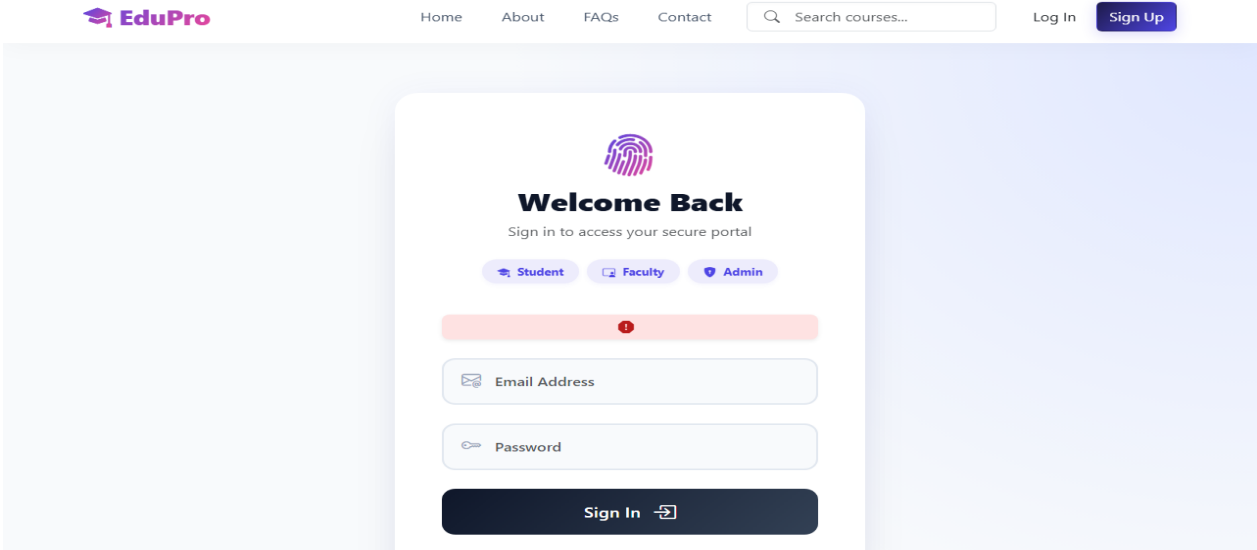


Figure 5: Notice creation interface supporting audience segmentation ('All', 'Student', 'Faculty'), urgency levels, and file attachment uploads.

Exhibit 6: Asynchronous Mass Email Broadcast Engine (/admin/broadcast)

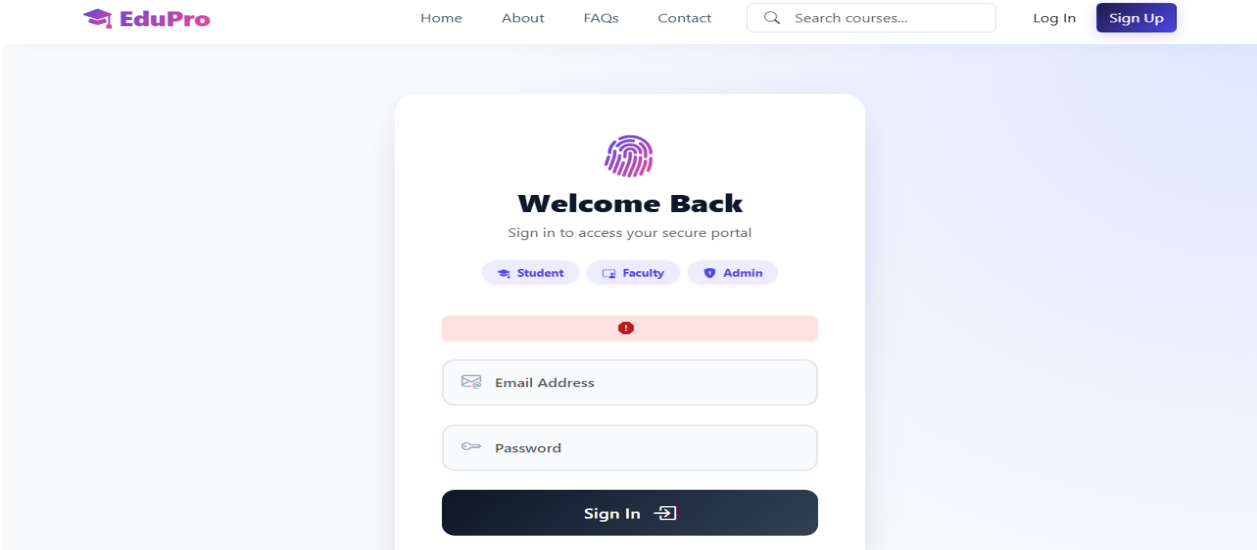


Figure 6: Rich-HTML broadcast dispatch tool with JavaMailSender async integration and complete immutable log audit table.

Exhibit 7: Faculty Academic Control Hub (/fdashboard)

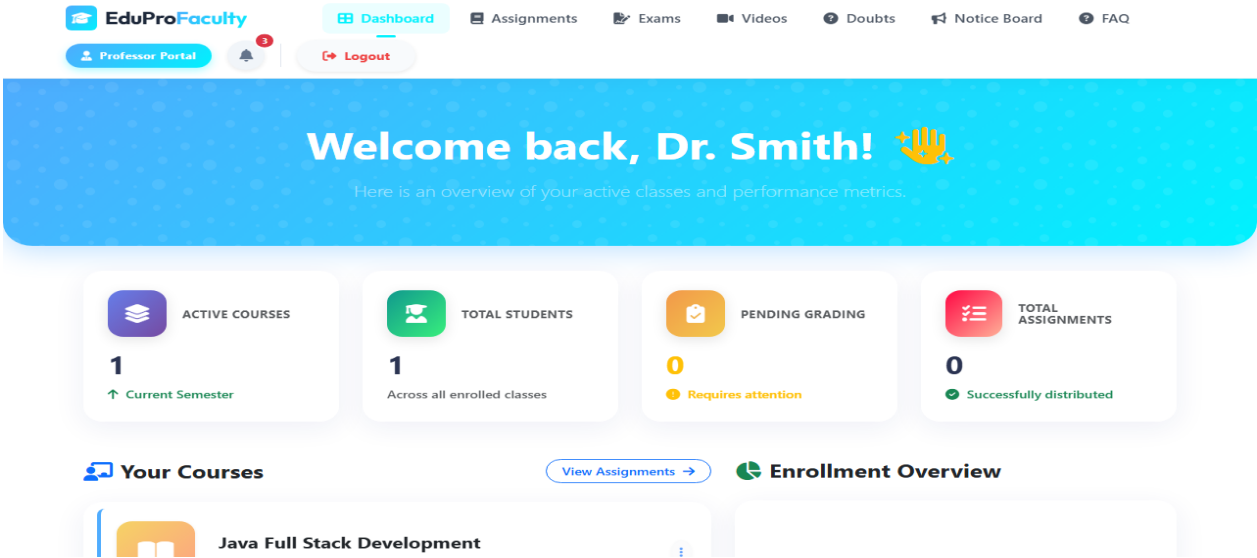


Figure 7: Instructor hub displaying active course stats, enrolled learner counts, pending assignment submissions, and quick action cards.

Exhibit 8: Assignment Authoring & Submissions Grading Portal (/f-assignments)

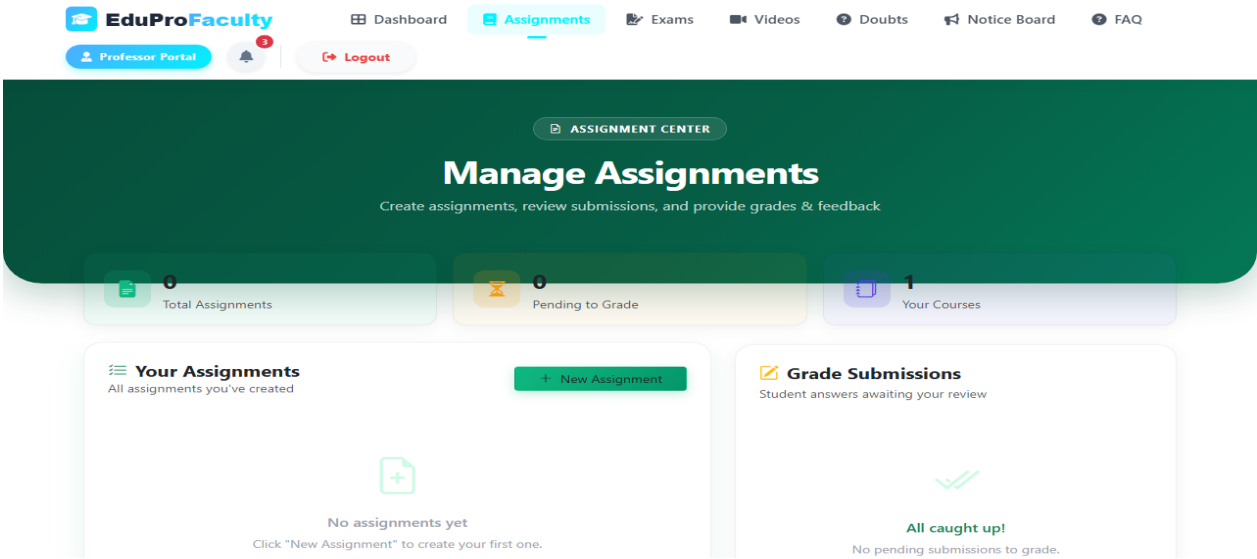


Figure 8: Faculty assessment interface allowing instructors to post homework briefs and grade uploaded student submissions with feedback.

Exhibit 9: Personalized Student Learning Center (/sdashboard)

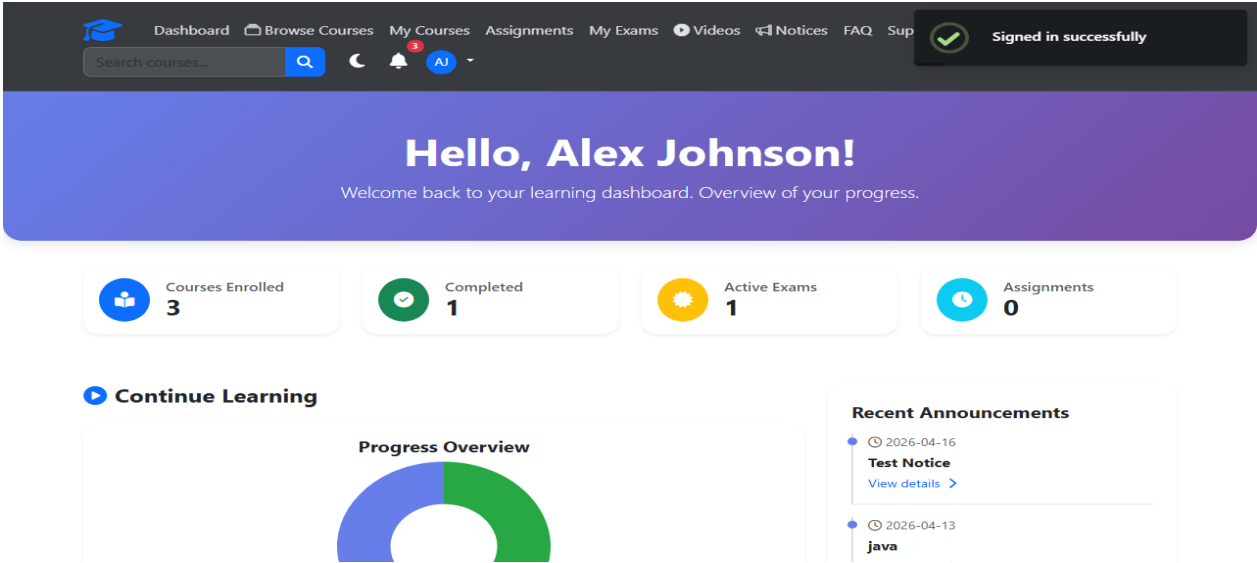


Figure 9: Student home showing active course progress bars, exam reminders, recent announcements, and doubt status badges.

Exhibit 10: Interactive Course Catalog & Video Player (/s-courses, /s-watch)

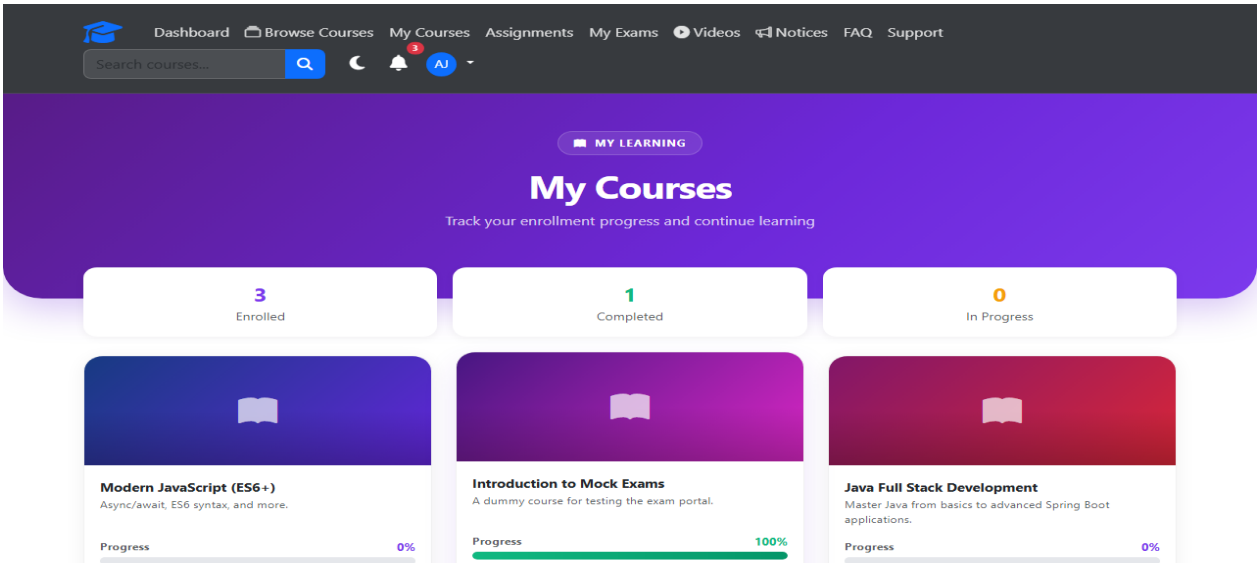


Figure 10: Video streaming interface with syllabus chapter lists, high-definition video playback, and dynamic completion tracking.

Exhibit 11: Student Assignment Submission & Grade Tracking (/s-assignments)

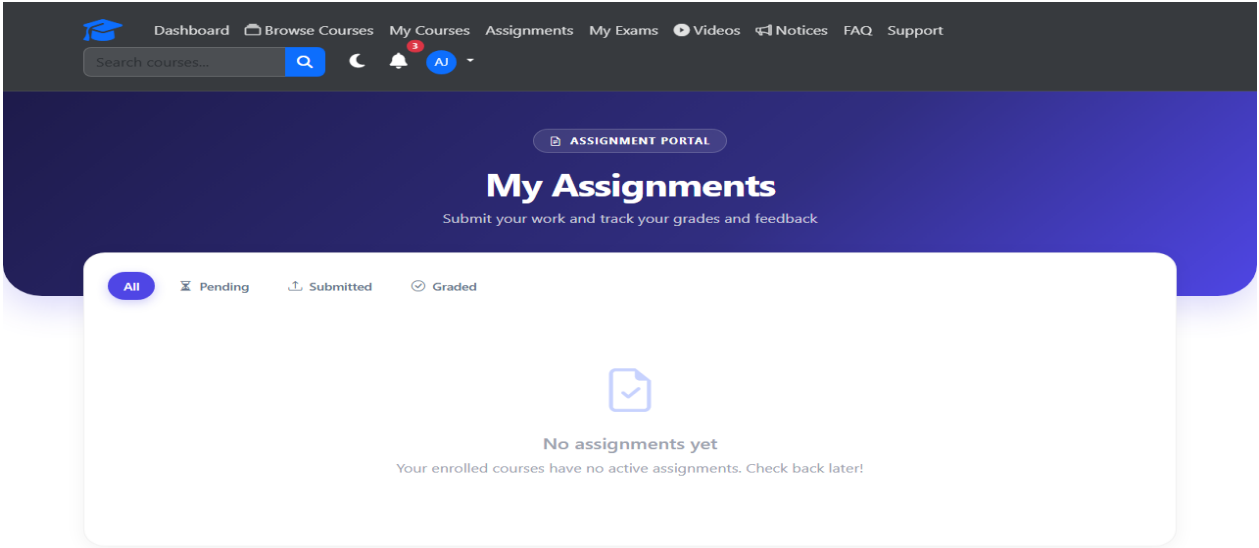


Figure 11: Learner submission portal displaying due dates, download links, file upload inputs, and instructor score reports.

Exhibit 12: Student Profile & Account Settings (/s-profile)

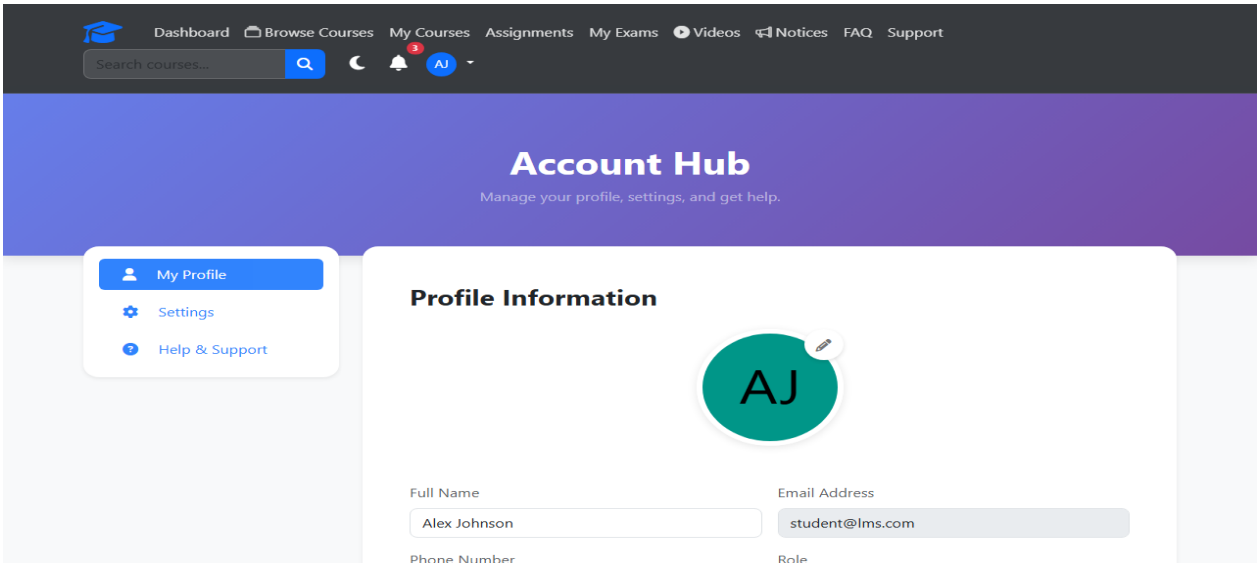


Figure 12: User management screen showing contact information, enrolled degree info, password update forms, and security history.

6. Controller & API Endpoint Specification

The table below outlines all primary HTTP endpoints exposed by Spring MVC controllers across the application.

Method & Endpoint	Controller Class	Access Role	Functional Description
GET / or /index	AnoController	Public	Serves the main landing page with course teasers and statistics.
GET /login	AnoController	Public	Renders the authentication login view.
POST /login	AnoController	Public	Authenticates credentials, checks status=1, sets session and isOnline=1.
GET /register	AnoController	Public	Renders new account registration view.
POST /register	AnoController	Public	Persists user with status=0 pending admin approval.
GET /logout	AnoController	Authenticated	Sets isOnline=0, invalidates HTTP session, redirects to /login.
GET /adashboard	AdminController	Admin	Renders Admin dashboard with calculated live KPI metrics.
GET /users	AdminController	Admin	Lists all users with activation toggles and role filters.
GET /edituser/{id}	AdminController	Admin	Renders edit user form for specified user ID.
POST /edituser	AdminController	Admin	Updates user details, status, and role assignment.
GET /deleteuser/{id}	AdminController	Admin	Permanently deletes user and associated records.
GET /addnotice	AdminController	Admin	Renders notice publication view with file upload.
POST /addnotice	AdminController	Admin	Saves notice, stores attachments, and publishes alert.
GET /admin/broadcast	BroadcastController	Admin	Renders mass Rich-HTML email broadcasting portal.
POST /admin/broadcast/send	BroadcastController	Admin	Dispatches asynchronous batch emails via JavaMailSender.
GET /admin/broadcast/logs	BroadcastController	Admin	Returns broadcast execution logs and status history.
GET /fdashboard	FacultyController	Faculty	Renders Faculty dashboard with course summaries.
GET /faculty/videos	VideoController	Faculty	Video management hub to upload and organize lectures.
POST /faculty/videos/upload	VideoController	Faculty	Handles multipart video upload (up to 500MB) with metadata.
GET /f-assignments	FacultyController	Faculty	Assignment creation and submission grading overview.
POST /faculty/grade-submission	FacultyController	Faculty	Records awarded marks and instructor feedback.
GET /sdashboard	StudentController	Student	Renders Student dashboard with progress trackers.
GET /s-courses	StudentController	Student	Displays enrolled courses and browsing catalog.
GET /s-watch	StudentController	Student	Streams course video lecture with interactive playlist.
GET /student/exams	StudentExamController	Student	Portal listing available, active timed examinations.
POST /student/exam/submit	StudentExamController	Student	Evaluates submitted answers, calculates score, and generates result.
GET /faq	AnoController	Public / All	Renders role-filtered Frequently Asked Questions.
POST /contact	AnoController	Public	Persists visitor inquiries and feedback into the database.

7. Installation, Configuration & Operations Guide

7.1. System Prerequisites

- **Java Development Kit (JDK):** Java 17 or higher (tested with Oracle JDK 24).
- **Database Server:** MySQL 8.0+ running on localhost:3306.
- **Build Management:** Apache Maven 3.8+ (or included Maven Wrapper `./mvnw`).
- **Operating System:** Cross-platform (Windows, Linux, macOS).

7.2. Database Configuration

Create the MySQL database using the MySQL Command Line Client or MySQL Workbench:

```
CREATE DATABASE lms;  
USE lms;
```

Configure database connection credentials in `src/main/resources/application.properties`:

```
server.port=8081  
spring.mvc.view.prefix=/views/  
spring.mvc.view.suffix=.jsp  
spring.datasource.url=jdbc:mysql://localhost:3306/lms  
spring.datasource.username=root  
spring.datasource.password=YOUR_MYSQL_PASSWORD  
spring.jpa.hibernate.ddl-auto=update  
spring.servlet.multipart.max-file-size=500MB  
spring.servlet.multipart.max-request-size=510MB
```

7.3. Building & Running the Application

Run the following Maven commands in the project root directory:

```
# Compile and build project  
./mvnw.cmd clean compile  
  
# Start Spring Boot server  
./mvnw.cmd spring-boot:run
```

Once started, access the application in any modern web browser at: **`http://localhost:8081`**

7.4. Default Credentials & Roles

Role	Email Address	Default Password	Landing Destination
Administrator	admin@lms.com	admin123	/adashboard (Admin Hub)
Faculty	faculty@lms.com	faculty123	/fdashboard (Faculty Portal)
Student	student@lms.com	student123	/sdashboard (Student Center)